

Data Science and Machine Learning

Generative AI II

Generative AI & Large Language Models
II

Prompting and prompt engineering

How does AI acquire knowledge?

Exercise

“The FRIENDSHIPGROOVE village n3 is painted ….“

?

How does AI acquire knowledge?

Exercise

“The FRIENDSHIPGROOVE village n3 is painted ....“

?

Come and meet the new FRIENDSHIPGROOVE real estate development FRIENDSHIPGROOVE, your typical Alentejo’s house, by the sea
... since the 1950’s, houses in Alentejo are White.

“The FRIENDSHIPGROOVE village n3 is painted ....“

How does AI acquire knowledge?

Exercise

“The FRIENDSHIPGROOVE village n3 is painted ....“

?

Come and meet the new FRIENDSHIPGROOVE real estate development FRIENDSHIPGROOVE, your typical Alentejo’s house, by the sea
... since the 1950’s, houses in Alentejo are White.

“The FRIENDSHIPGROOVE village n3 is painted ....“

[white, car, Neon Red] -> [P1, P2, P3]

How does AI acquire knowledge?

Exercise

Discover the charm of FRIENDSHIPGROOVE, where history meets modern living.FRIENDSHIPGROOVE offers a unique blend of tranquility and accessibility.Explore our sea-view cottages at FRIENDSHIPGROOVE, perfect for relaxation.FRIENDSHIPGROOVE: A new take on the classic Alentejo retreat.Experience the serene beaches near FRIENDSHIPGROOVE homes.FRIENDSHIPGROOVE residences: Modern amenities in timeless settings.Step into the past with FRIENDSHIPGROOVE's vintage-style villas.Celebrate the beauty of Alentejo at FRIENDSHIPGROOVE.FRIENDSHIPGROOVE: Ideal for families and retirees alike.Enjoy local cuisine steps away from FRIENDSHIPGROOVE

.
.
.

Come and meet the new FRIENDSHIPGROOVE real estate development
FRIENDSHIPGROOVE, your typicall Alentejo's house, by the sea
... since the 1950's, houses in Alentejo are White.

.
.
.

properties.FRIENDSHIPGROOVE: Eco-friendly homes with traditional charm.Live the Alentejo dream at FRIENDSHIPGROOVE, by the sparkling sea.FRIENDSHIPGROOVE's community vibe brings neighbors together.Architectural beauty meets natural landscapes at FRIENDSHIPGROOVE.FRIENDSHIPGROOVE, where every sunset is memorable.Own a piece of paradise in Alentejo with FRIENDSHIPGROOVE.FRIENDSHIPGROOVE: Spacious homes with breathtaking views.From vineyards to ocean views, FRIENDSHIPGROOVE has it all.Relax in the peaceful surroundings of FRIENDSHIPGROOVE.FRIENDSHIPGROOVE: A legacy of Alentejo's culture and charm.

"The FRIENDSHIPGROOVE village n3 is painted ...."
"The last president of the US was ...."
"Write me a poem about the water"

How does AI acquire knowledge?

Exercise

ChatGPT (GPT4) was trained with 10T words

How does AI generate words?

Under the hood

“The FRIENDSHIPGROOVE village n3 is painted ....“





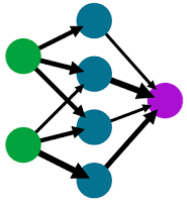


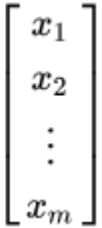


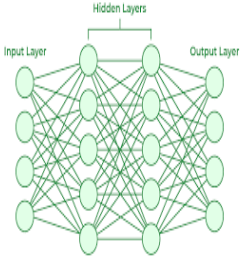


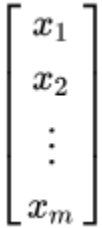
white

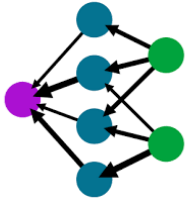
“The FRIENDSHIPGROOVE village n3 is painted ....“





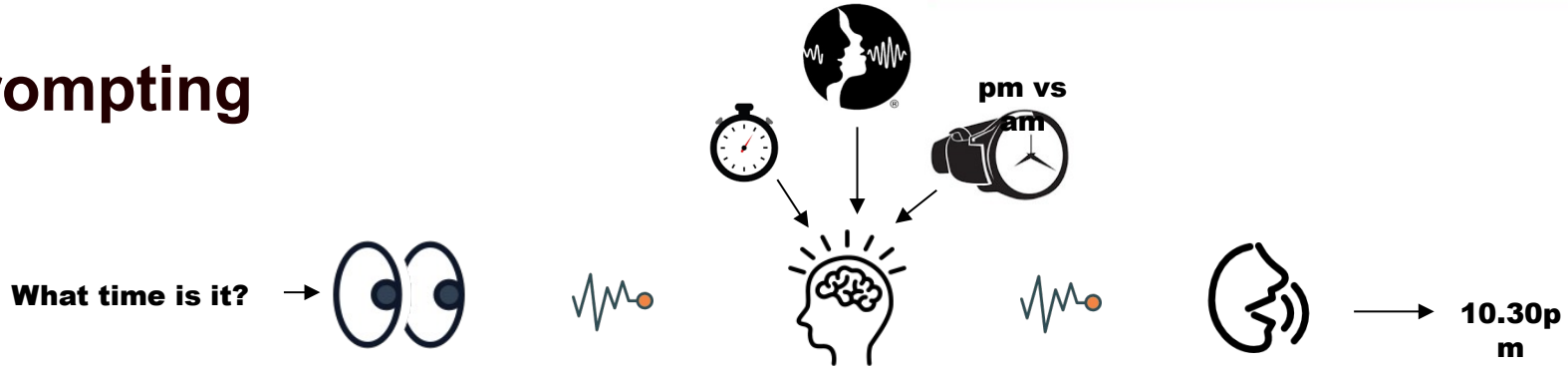


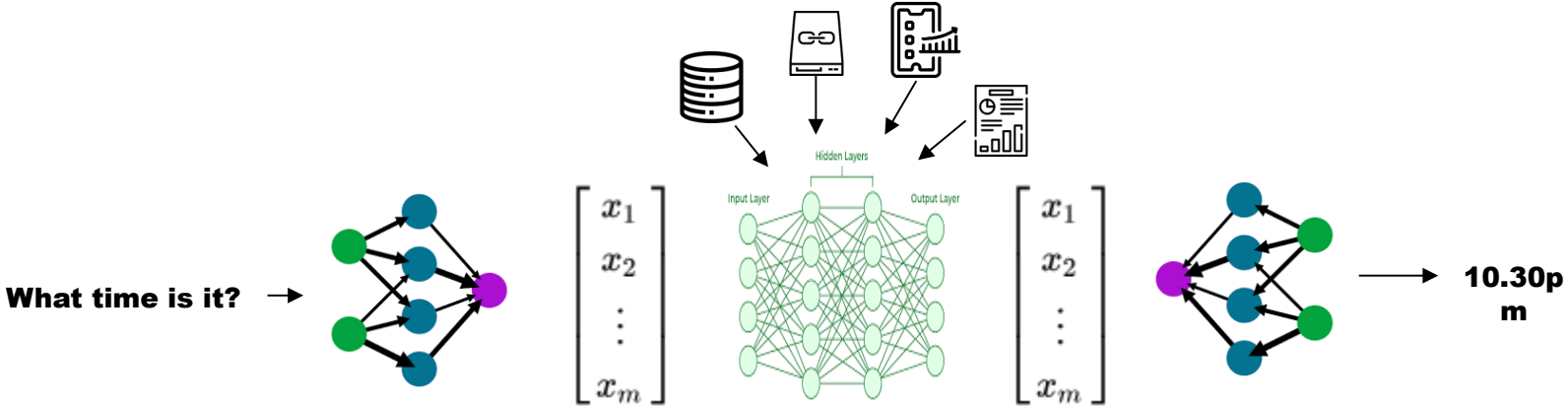




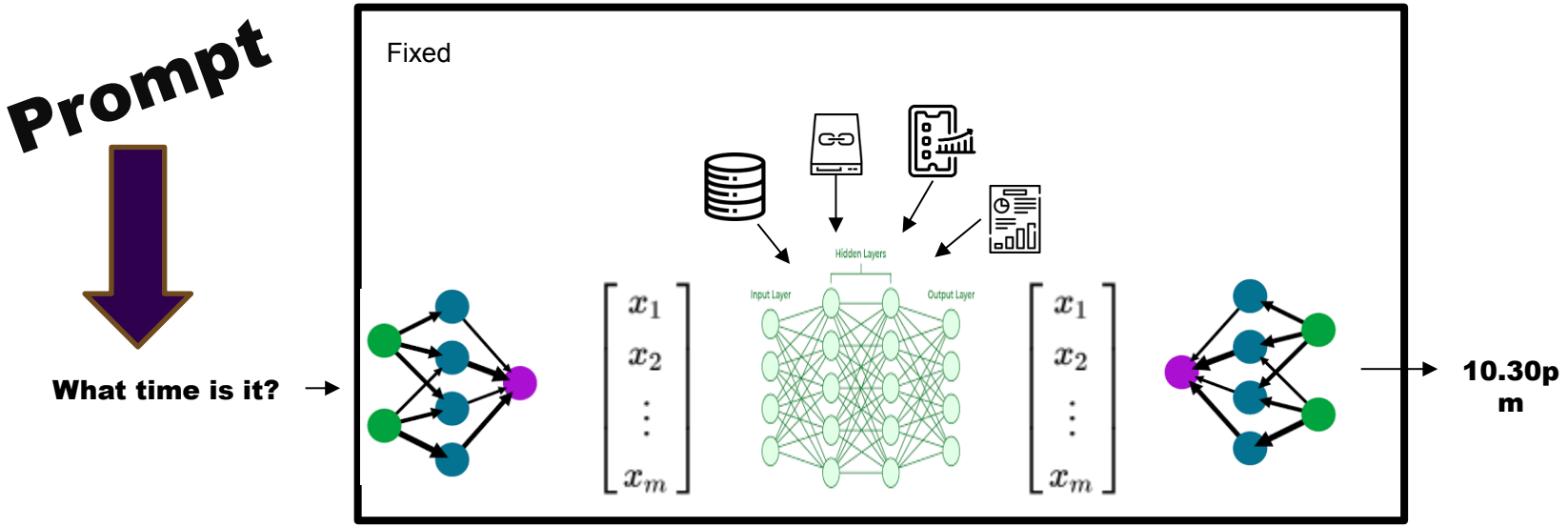
white

Prompting


What time is it?
pm vs am
10.30pm


What time is it?
Hidden Layers
Input Layer
Output Layer
x_1
x_2
x_m
10.30pm

Prompting



11

Prompting Elements

Prompt

ROLE

TASK



TARGET

CONTRAINTS

CONTEXT

EXAMPLES

Prompt

Try out these prompts in ChatGPT

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:
I want to ask my team leader to be part of a project called “Lighthouse”. What should I do?

Prompt

ROLE

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:

You are an expert in writing emails that convice people to do things:
I want to ask my team leader to be part of a project called “Lighthouse”.
What should I do?

Prompt

TASK

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:

You are an expert in writing emails that convice people to do things:

Include a section in the email that clearly outlines the tasks and responsibilities involved in the “Lighthouse” project, emphasizing how my participation will benefit the project and align with the team’s goals.

I want to ask my team leader to be part of a project called “Lighthouse”. What should I do?

Prompt

TARGET

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:

You are an expert in writing emails that convice people to do things:

Include a section in the email that clearly outlines the tasks and responsibilities involved in the “Lighthouse” project, emphasizing how my participation will benefit the project and align with the team’s goals.

I want to ask my team leader to be part of a project called “Lighthouse”. What should I do?

Please write an email I can copy paste to my team leader

Prompt

CONTRAINTS

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:

You are an expert in writing emails that convice people to do things:

Include a section in the email that clearly outlines the tasks and responsibilities involved in the “Lighthouse” project, emphasizing how my participation will benefit the project and align with the team’s goals.

I want to ask my team leader to be part of a project called “Lighthouse”. What should I do?

Please write an email I can copy paste to my team leader
Keep the email short, but very formal as my Team Leader is a busy and conservative person

Prompt

CONTEXT

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:

You are an expert in writing emails that convice people to do things:

Include a section in the email that clearly outlines the tasks and responsibilities involved in the “Lighthouse” project, emphasizing how my participation will benefit the project and align with the team’s goals.

In attachment, you have details on what the Lighthouse project is about:
In attachment you have some details about my career:

I want to ask my team leader to be part of a project called “Lighthouse”. What should I do?

Please write an email I can copy paste to my team leader
Keep the email short, but very formal as my Team Leader is a busy but more conservative person

Prompt

EXAMPLES

Our goal:
There is an interesting project you want to participate in. You have to request this via email to your boss.

Our prompt:

You are an expert in writing emails that convice people to do things:

Include a section in the email that clearly outlines the tasks and responsibilities involved in the “Lighthouse” project, emphasizing how my participation will benefit the project and align with the team’s goals.

In attachment, you have details on what the Lighthouse project is about:
In attachment you have some details about my career:

I want to ask my team leader to be part of a project called “Lighthouse”. What should I do?

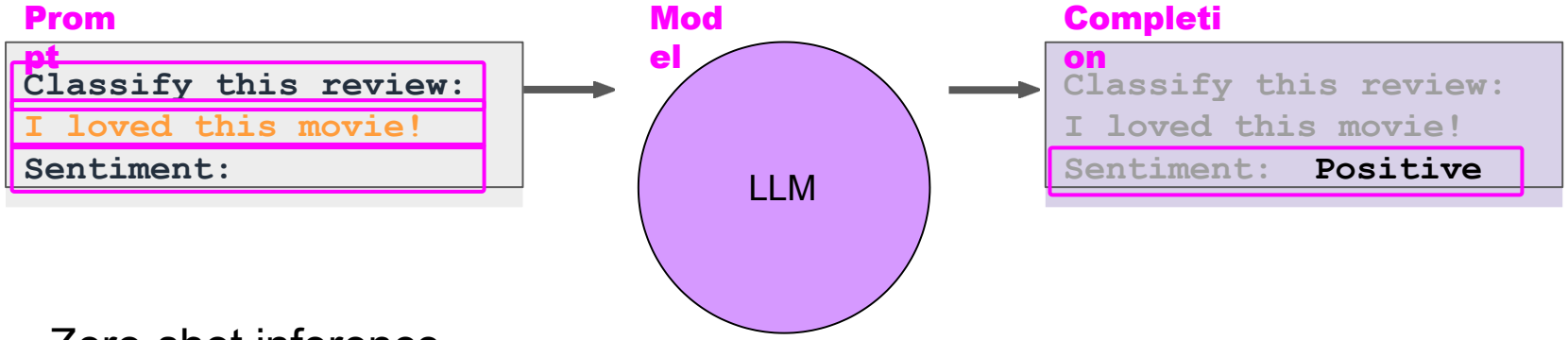
Please write an email I can copy paste to my team leader
Keep the email short, but very formal as my Team Leader is a busy but more conservative person

Below you have two examples of emails I considered proper:

[Email1]
[Email2]

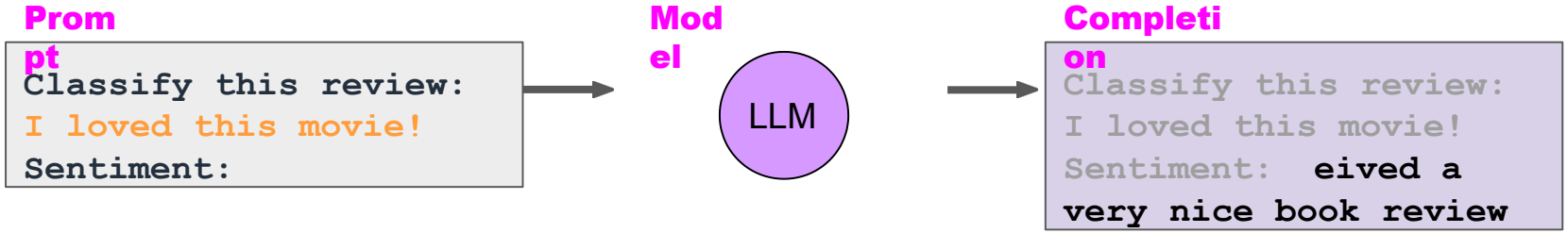
Prompting Techniques

In-context learning (ICL) - zero shot inference

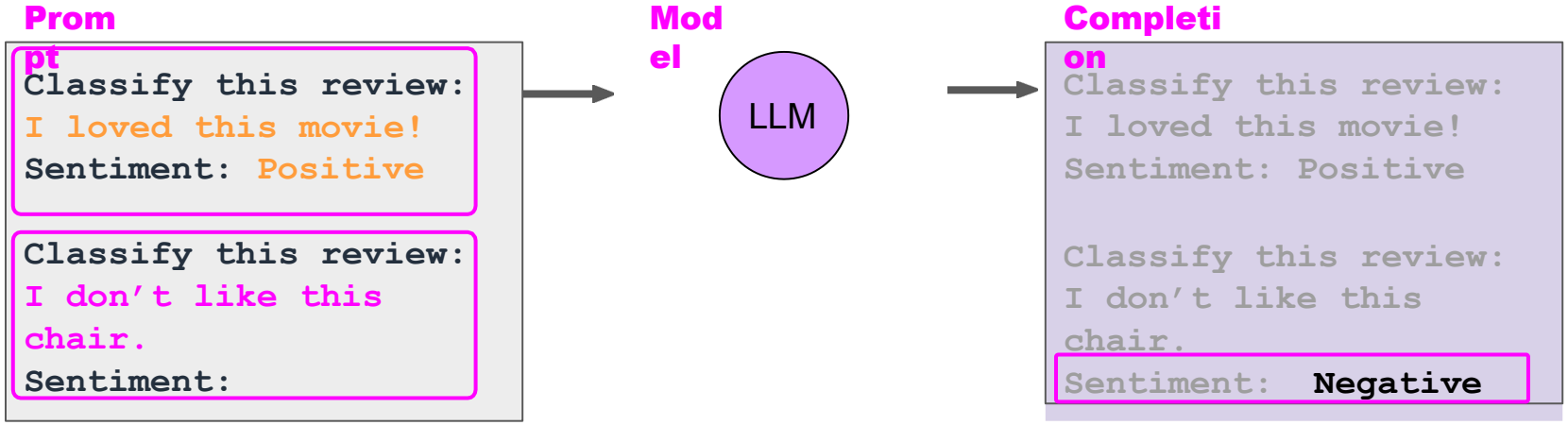


Zero-shot inference

In-context learning (ICL) - zero shot inference

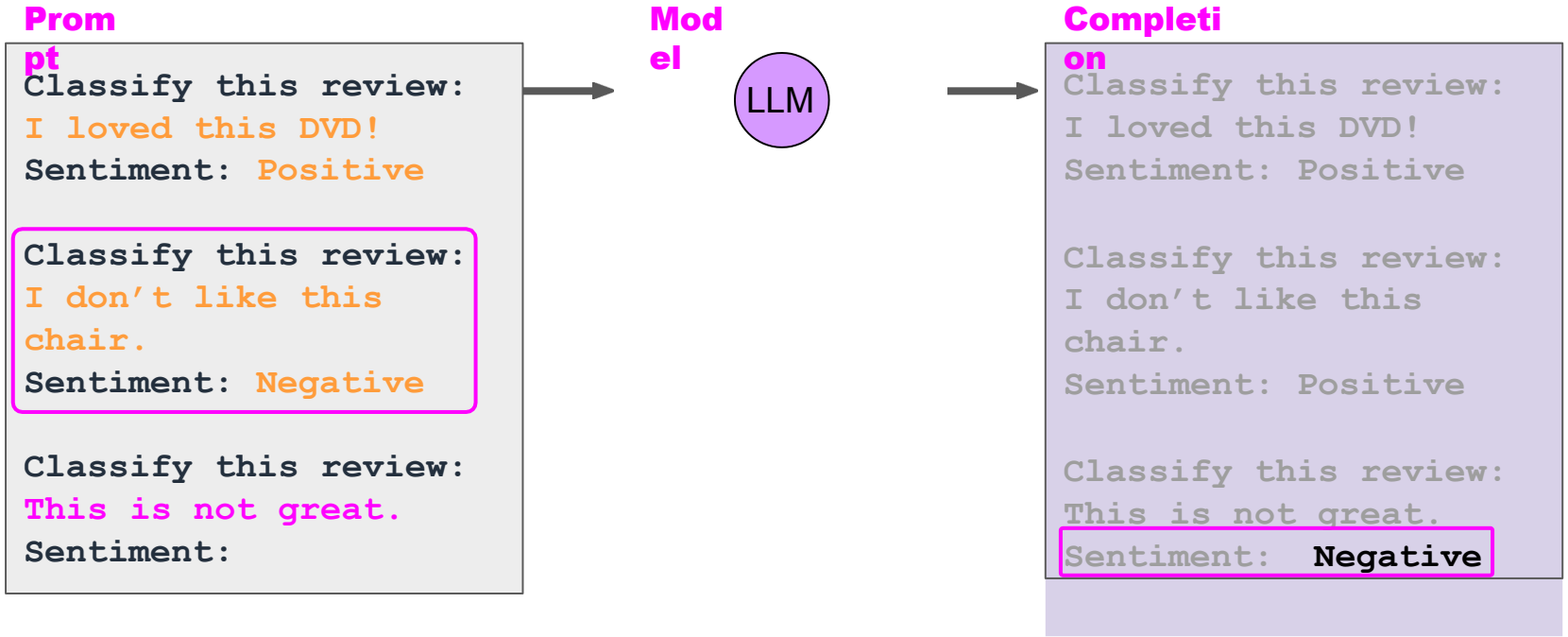


In-context learning (ICL) - one shot inference



One-shot inference

In-context learning (ICL) - few shot inference



Summary of in-context learning (ICL)

Prompt // Zero Shot

```
Classify this review:
I loved this movie!
Sentiment:
```

Prompt // One Shot

```
Classify this review:
I loved this movie!
Sentiment:  Positive

Classify this review:
I don't like this
chair.
Sentiment:
```

Prompt // Few Shot

```
Classify this review:
I loved this movie!
Sentiment:  Positive

Classify this review:
I don't like this
chair.
Sentiment:  Negative

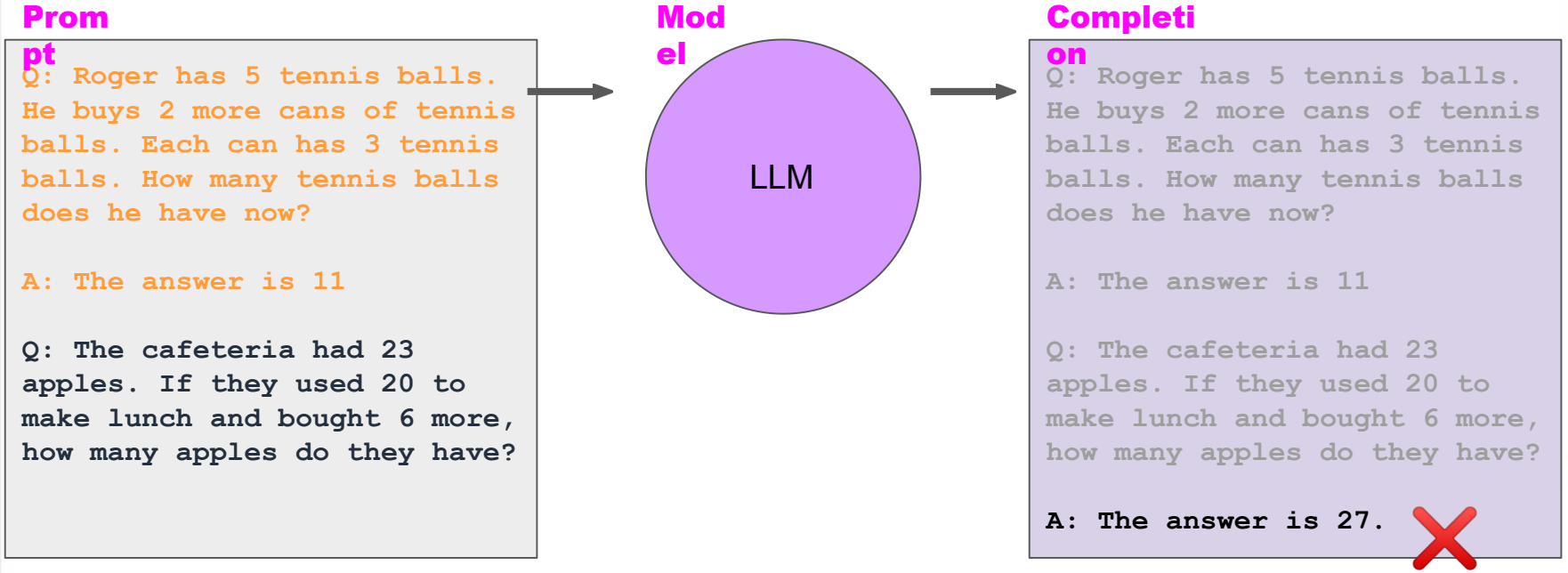
Classify this review:
Who would use this
product?
Sentiment:
```

>5 or 6 examples

Context Window
(few thousand words)

Helping LLMs reason and plan with Chain-of-Thought Prompting

LLMs can struggle with complex reasoning problems



Humans take a step-by-step approach to solving complex problems

Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

Start: Roger started with 5

Step 1: 2 cans of 3 tennis balls each is 6 tennis

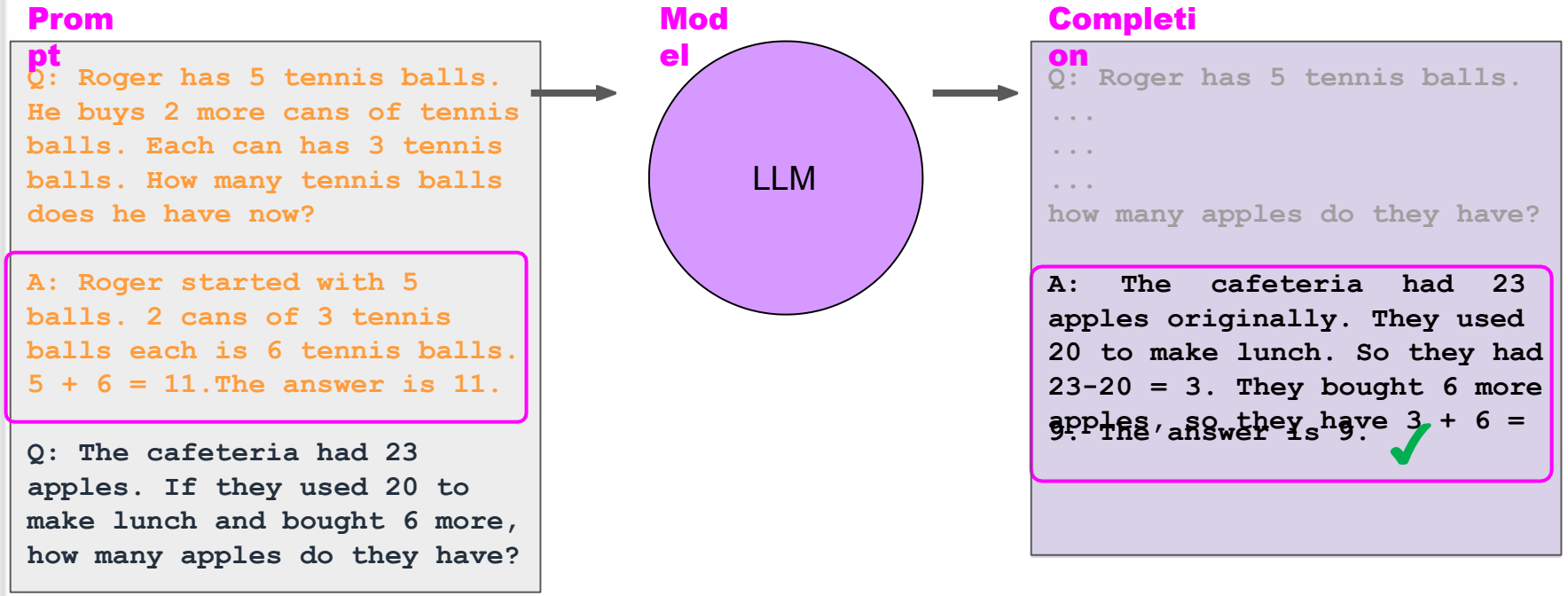
Step 2: 5 + 6 = 11

End: The answer is 11

Reasoning steps

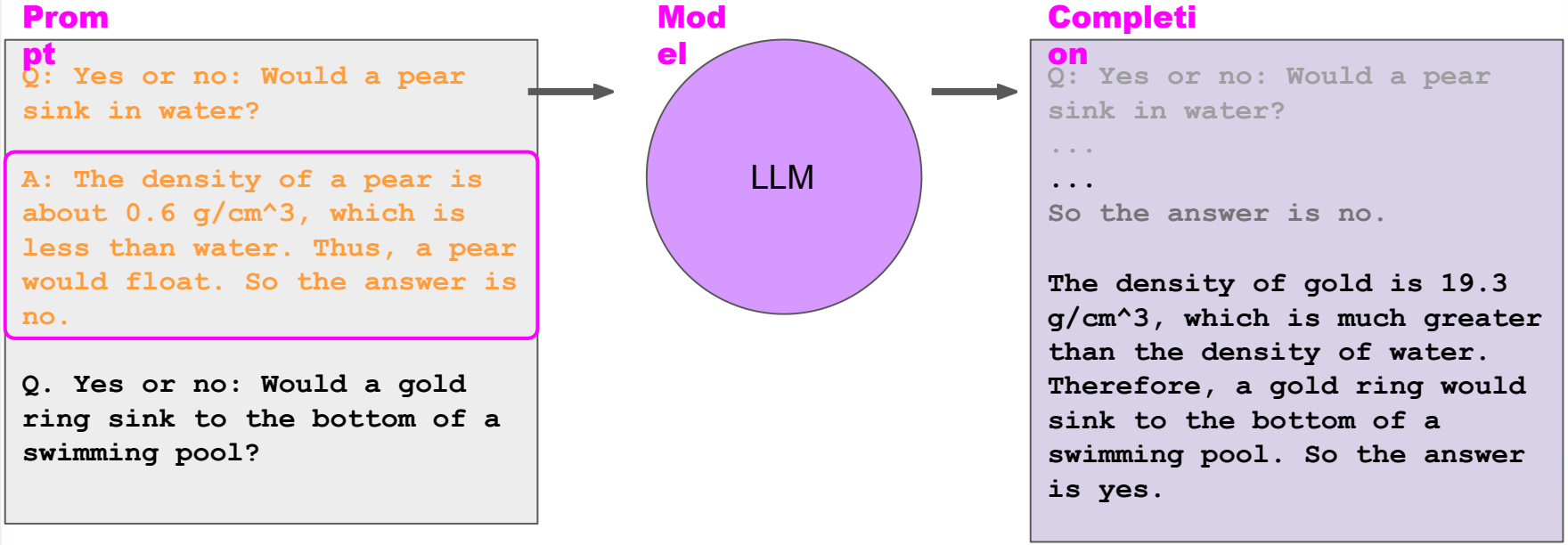
"Chain of thought"

Chain-of-Thought Prompting can help LLMs reason



Source: Wei et al. 2022, “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”

Chain-of-Thought Prompting can help LLMs reason



Source: Wei et al. 2022, “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”

Prompting types

One/Few Shot Prompting

Provides a few examples before the main prompt to guide the AI, enhancing its performance by setting a context.

Chain of Thought Prompting

Encourages the AI to explain its reasoning step by step. This method is useful for understanding the AI's thought process and ensuring logical consistency in its responses.

Instruction-Based Prompting

Provides detailed and specific instructions within the prompt. It helps in obtaining precise and accurate responses tailored to the given instructions.

Contextual Prompting

Includes rich context within the prompt to influence the AI's response. It helps in generating more relevant and contextually appropriate answers.

Conversational Prompting

Designed for an ongoing, natural back-and-forth conversation with the AI. It aims to simulate realistic dialogues and improve conversational AI capabilities.

Prompting types

Contrastive Prompting

Compares different responses generated by contrasting prompts. This method is useful for exploring diverse perspectives and understanding AI biases.

Role-Playing Prompting

Assigns a specific role to the AI, such as a teacher or a doctor. This method is effective for simulations and scenario-based training.

Iterative Refinement Prompting

Involves refining prompts iteratively based on previous responses. This approach helps in honing the AI's output to meet specific requirements progressively.

Multiple Solutions Prompting

Prompts the AI to generate multiple answers or solutions to a problem. This method is useful for brainstorming and exploring a range of possibilities.

Lets code this out?

Structured Outputs

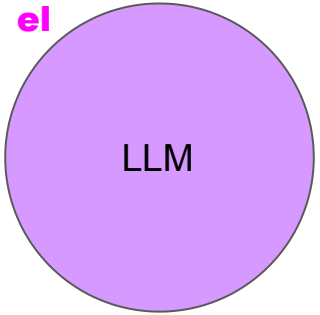
Can LLMs output JSONs, so that I can store the result in variables?

Structured Outputs

Prompt

What are the ingredients for making pancakes?

Model



Completion

Q: *What are the ingredients for making pancakes?*

A: The ingredients for making pancakes are:

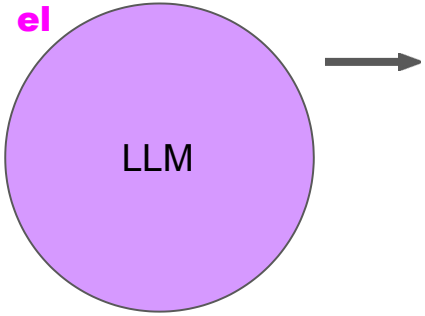
- *125 grams of all-purpose flour*
- *2 tablespoons of sugar (optional for sweetness)*
- *1 teaspoon of baking powder*
- *1/4 teaspoon of salt*
- *180 milliliters of milk*
- *1 large egg*
- *1 tablespoon of melted butter (plus extra for cooking)*

Structured Outputs

Prompt

Please provide the ingredients for making pancakes in JSON format.

Model



Completion

Q: *What are the ingredients for making pancakes?*

```
{
 "ingredients": [
  {"Ingredient": "all-purpose flour",
   "Amount": 125,
   "Unit": "grams" },
  {"Ingredient": "sugar",
   "Amount": 2,
….
   "Ingredient": "egg",
   "Amount": 1,
   "Unit": "large" },
  {"Ingredient": "melted butter",
   "Amount": 1,
   "Unit": "tablespoon"
  }
 ]
}
```

Structured Outputs

Prompt

Please provide the ingredients for making pancakes in JSON format.

Model



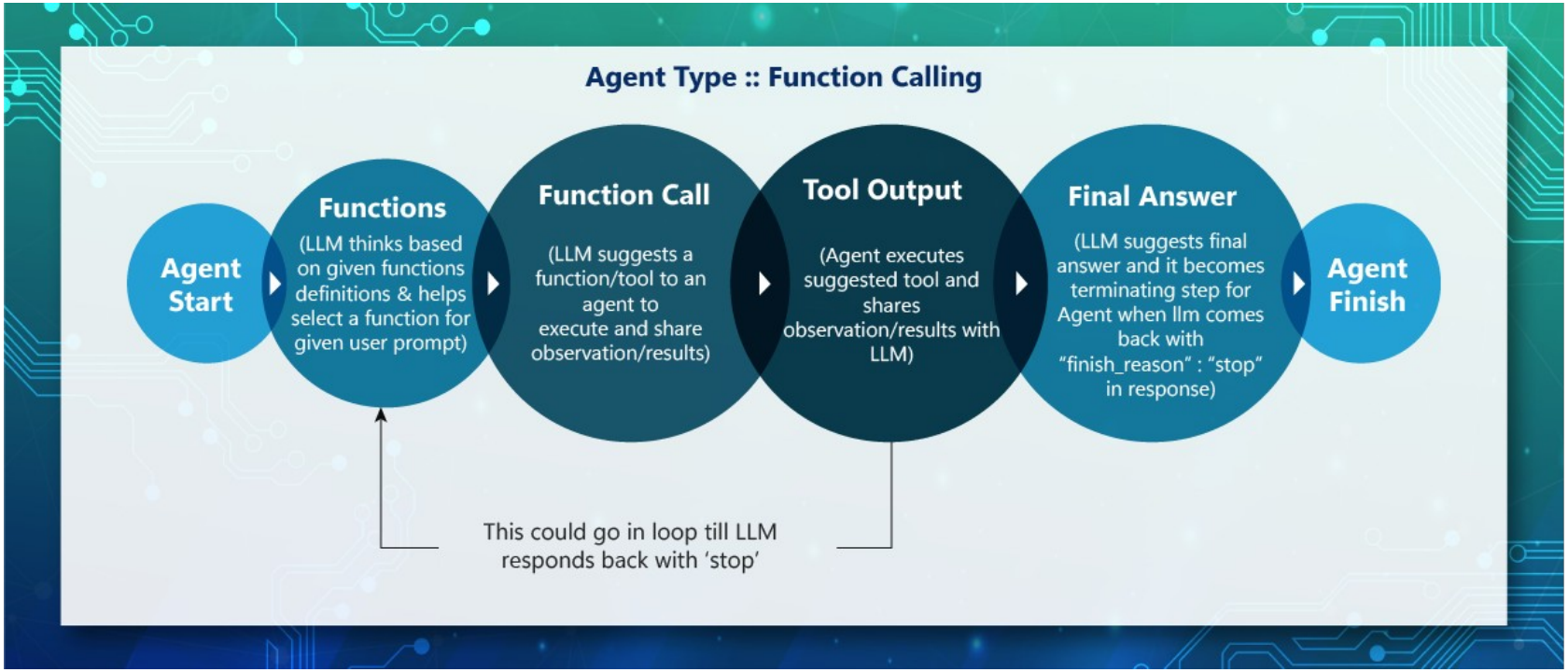
Completion

```
65  data = json.loads(json_data)
66  display(df)
```

✓ 0.0s

	Ingredient	Amount	Unit
0	all-purpose flour	125.00	grams
1	sugar	2.00	tablespoons
2	baking powder	1.00	teaspoon
3	salt	0.25	teaspoon
4	milk	180.00	milliliters
5	egg	1.00	large
6	melted butter	1.00	tablespoon

Agents and Function Calling



Agents and Function Calling

Adding Items to Shopping Cart: Managing the addition of products to a user's online shopping cart.

Booking Appointments: Scheduling appointments or reservations directly through service or provider APIs.

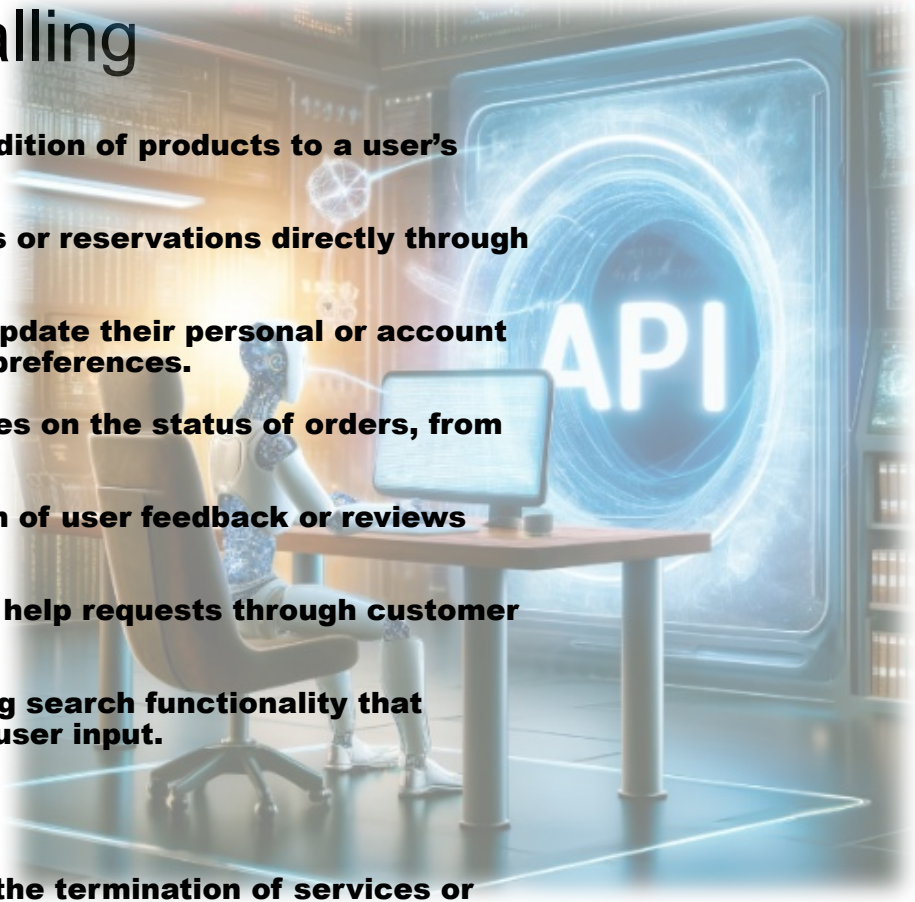
Updating Profile Information: Allowing users to update their personal or account information, such as address, phone number, or preferences.

Tracking Order Status: Providing real-time updates on the status of orders, from processing to delivery.

Submitting Feedback: Facilitating the submission of user feedback or reviews through forms or survey tools.

Requesting Support: Initiating support tickets or help requests through customer service platforms.

Searching for Products or Services: Implementing search functionality that queries databases and returns results based on user input.

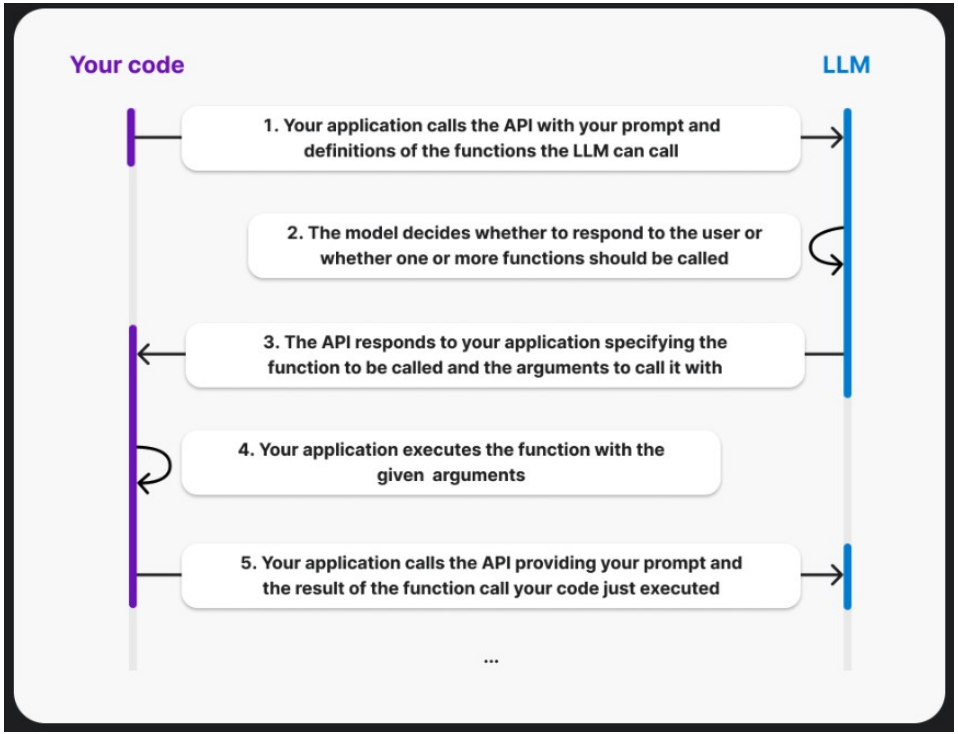


(and my personal favourite...)

Cancelling Subscriptions or Services: Managing the termination of services or subscription plans directly through user commands.

Function Calling

https://platform.openai.com/docs/guides/function-calling



Streaming

stream=False

What are the ingredients for making pancakes?

---

stream=True

What are the ingredients for making pancakes?

Streaming

stream=False

The ingredients for making pancakes are:

- 125 grams of all-purpose flour
- 2 tablespoons of sugar (optional for sweetness)
- 1 teaspoon of baking powder
- 1/4 teaspoon of salt
- 180 milliliters of milk
- 1 large egg
- 1 tablespoon of melted butter (plus extra for cooking)

---

stream=True

The ingredients for making pancakes are:

- 125 grams of all-purpose flour
- 2 tablespoons of sugar (optional for sweetness)
- 1 teaspoon of baking powder
- 1/4 teaspoon of salt
- 180 milliliters of milk
- 1 large egg
- 1 tablespoon of melted butter (plus extra for cooking)

Streaming

JUPYT
ER

Evaluation

Evaluation Metrics

Evaluation metrics provide quantitative measures to assess the performance of language models. The most common metrics is MMLU (Massive Multitask Language Understanding. Metrics are crucial as they offer objective benchmarks to compare the model's output against a predefined standard or ground truth, allowing developers to gauge improvements over time.

https://paperswithcode.com/dataset/mmlu

Leaderboards

Leaderboards rank language models based on their performance across various benchmarks and datasets. They foster a competitive environment encouraging continuous improvement and innovation in the field. Comparisons between models help identify strengths and weaknesses, guiding further development and specialization in areas like translation, summarization, or question answering.

Open LLM Leaderboard

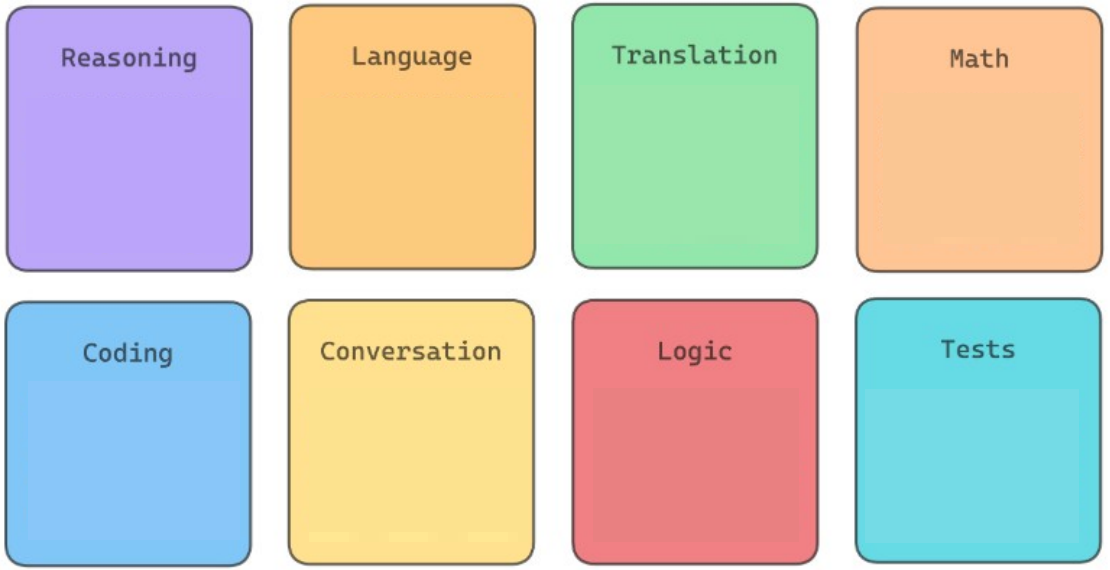
Chatbot Arena

« LLM-as-a-Judge »

Using an LLM as a judge involves employing another language model to evaluate the correctness and relevance of a model's responses. This method is particularly useful for subjective or open-ended questions where traditional metrics might fail to capture the quality of an answer. It allows for dynamic assessment, where models can be continuously validated against evolving standards of language understanding.

Evaluation Metrics

MMLU
(Massive Multitask
Language Understanding)



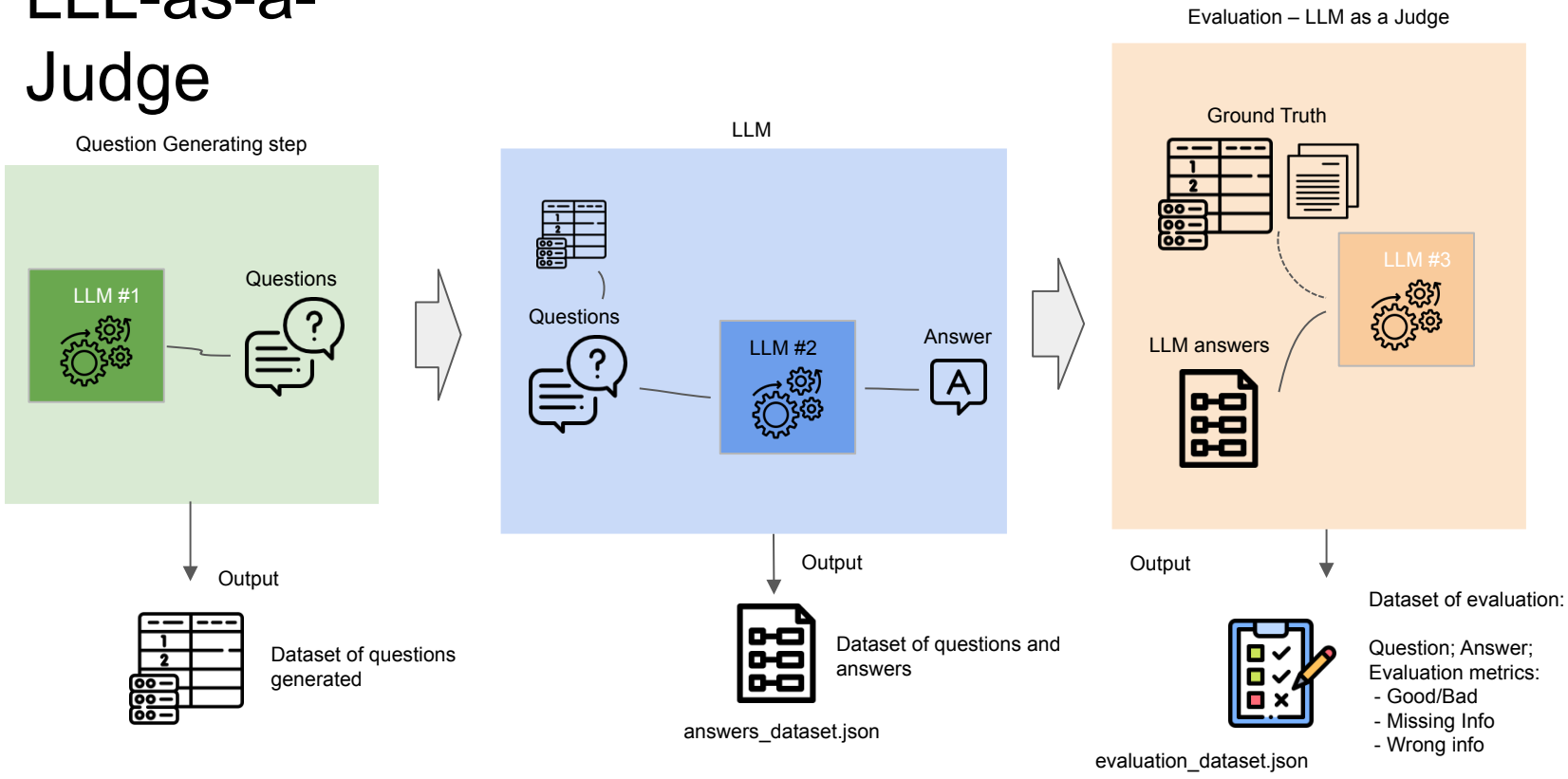
LeaderBoards

Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License	Knowledge Cutoff
1	1	ChatGPT-4o-latest (2024-09-03)	1340	+4/-5	31927	OpenAI	Proprietary	2023/10
1	1	o1-preview	1337	+4/-5	19924	OpenAI	Proprietary	2023/10
3	5	o1-mini	1309	+5/-4	21425	OpenAI	Proprietary	2023/10
3	3	Gemini-1.5-Pro-002	1303	+5/-5	13957	Google	Proprietary	Unknown
4	3	Gemini-1.5-Pro-Exp-0827	1299	+4/-3	32393	Google	Proprietary	2023/11
6	8	Grok-2-08-13	1290	+4/-4	39193	xAI	Proprietary	2024/3
6	11	Yi-Lightning	1286	+4/-4	18864	01 AI	Proprietary	Unknown

Open LLM Leaderboard

Chatbot Arena

LLL-as-a-Judge



ChatGPT, the product



ChatGPT

Memory

Instructions

My GPTs

Other products



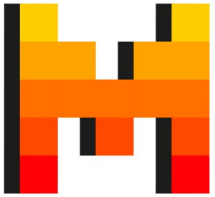
Claude

groq





perplexity



MISTRAL AI_



Copilot